

RAZVAN MIHAI POPESCU

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EDUCATION

Delft University of Technology

Ph.D. in Computer Science, AI-enabled SE

Supervisors: Maliheh Izadi, Arie van Deursen

Delft, NL

Mar. 2025 - Present

Delft University of Technology

Master of Computer Science, Artificial Intelligence

Delft, NL

Sept. 2023 - Feb. 2025

Courses: Supercomputing for Big Data, Software Architecture, Information Retrieval, Artificial Intelligence Techniques, Conversational Agents, Analytics and Machine Learning for Software Engineering, Testing and Validation for AI-Intensive Systems, Artificial Intelligence for Software Testing and Reverse Engineering, Natural Language Processing, Release Engineering for Machine Learning Applications

Delft University of Technology

Bachelor of Computer Science and Engineering

Delft, NL

Sept. 2020 - Jul. 2023

Courses: Algorithms and Data Structures, Object-oriented Programming, Software Engineering Methods, Software Quality and Testing, Big Data Processing, Data Mining, Machine Learning, Computational Intelligence, Collaborative Artificial Intelligence, Human-Computer Interaction

PUBLICATIONS

Beyond Merging: Refactoring and Code Quality Signals in Agent-Authored Pull Requests

Under Review

Inaesh Joshi, Razvan Mihai Popescu, Maliheh Izadi

Characterizing Human-Agent Dynamics in Agent-authored Pull Requests

Under Review

Ziyou Li, Razvan Mihai Popescu, Maliheh Izadi

When the Docstring Lies, LLM Judges Blame the Code

Under Review

Razvan Mihai Popescu, Maliheh Izadi

Evaluating Large Language Models in Code-Related Tasks: A Systematic Literature Review

Under Review

Razvan Mihai Popescu, Maliheh Izadi

Reliable and Developer-Aligned Evaluation of Agents for Software Engineering

FSE '26

Razvan Mihai Popescu

The Poisoned Chalice of LLM Evaluation Report

FSE '26

Jonathan Katzy, Ali Al-Kaswan, Razvan Mihai Popescu, Zhou Yang

Investigating Autonomous Agent Contributions in the Wild: Activity Patterns and Code Change over Time

MSR '26

Razvan Mihai Popescu, David Gros, Andrei Botocan, Rahul Pandita, Prem Devanbu, Maliheh Izadi

Silent Reliance or Diligent Review? Human-Agent Engagement and Interaction Patterns in GitHub

ICSE '26

Ziyou Li, Razvan Mihai Popescu, Maliheh Izadi

Automated Attention Pattern Discovery at Scale in Large Language Models (extension)

TMLR '26

Jonathan Katzy, Razvan Mihai Popescu, Erik Mekkes, Arie van Deursen, Maliheh Izadi

Automated Attention Pattern Discovery at Scale

NeurIPS '25

Jonathan Katzy, Razvan Mihai Popescu, Erik Mekkes, Arie van Deursen, Maliheh Izadi

The Heap: A Contamination-Free Multilingual Code Dataset for Evaluating Large Language Models

FORGE '25

Jonathan Katzy, Razvan Mihai Popescu, Arie van Deursen, Maliheh Izadi

An Exploratory Investigation into Code License Infringements in LLM Training Datasets

FORGE '24

Jonathan Katzy, Razvan Mihai Popescu, Arie van Deursen, Maliheh Izadi

Language Models for Code Completion: A Practical Evaluation

ICSE '24

Maliheh Izadi, Jonathan Katzy, Tim van Dam, Marc Otten, Razvan Mihai Popescu, Arie van Deursen

A Study on the Impact of Common Code Structures on CodeParrot's Autocompletion Performance

TU Delft

Razvan Mihai Popescu

RESEARCH EXPERIENCE

Doctoral Researcher

Mar. 2025 – Present

AISE Lab

TU Delft, NL

- Analyzed 5M agent and human contributions, finding that agents refactor less and exhibit more code smells
- Evaluated 50K verdicts from five LLM judges, showing false docstrings cut acceptance of correct code by up to 82%
- Tracked 110K agent and human contributions, revealing lower survival and higher churn for agent-authored code
- Identified limited human oversight of coding agents, with 74% of 212K agent pull requests merged without modification
- Proposed a doctoral research plan for developer-aligned, real-world evaluation of LLM-powered coding agents
- Conducted a systematic literature review on evaluation techniques and experimental settings for LLMs in SE

Research Assistant

Mar. 2023 – Mar. 2025

AISE Lab

TU Delft, NL

- Co-developed AP-MAE, a vision transformer for scalable LLM attention analysis, achieving a 13.6% accuracy gain
- Built The Heap, a contamination-free code corpus spanning 57 programming languages for fair LLM evaluation
- Audited public LLM training datasets for license compliance, revealing infringement rates of up to 22%
- Contributed to Code4Me, an open-source IDE extension for evaluating LLM code completion in real-world settings
- Showed that common code structures reduce CodeParrot's null attention by 30% and prediction depth by 50%

TALKS & PRESENTATIONS

- *Reliable and Developer-Aligned Evaluation of Agents for Software Engineering*, FSE '26, Montreal, Canada
- *Investigating Autonomous Agent Contributions in the Wild*, MSR '26, Rio de Janeiro, Brazil
- *The Heap: A Contamination-Free Multilingual Code Dataset*, FORGE '25, Ottawa, Canada

ACADEMIC SERVICE

- Co-chair of the [Membership Inference Attack Competition](#) at FSE '26 and ICSE '27
- Web chair of the [AgenticSE Workshop](#) at ASE '25
- Reviewer for ICSE, FSE, ASE, ISSTA, and TOSEM

TEACHING & SUPERVISION

Thesis Supervisor

Mar. 2025 – Present

AISE Lab

TU Delft, NL

- Supervising Nawmi Nujhat's MSc thesis on agentic issue resolution within dynamic software environments
- Supervised Inaesh Joshi's MSc thesis on agentic refactoring with Meta, leading to the paper [Beyond Merging](#)
- Supervised 4 BSc students on how data smells such as boilerplate, non-English code, and technical debt affect LLMs

Teaching Assistant

Feb. 2025 – Apr. 2025

Responsible Data Science and AI Engineering

TU Delft, NL

- Facilitated in-class discussions on the ethical and societal implications of AI systems in a 400-student course
- Graded student reports and final projects on the design of responsible AI solutions
- Co-organized and assessed final project presentations, providing feedback to student teams

Teaching Assistant

Nov. 2024 – Feb. 2025

Machine Learning for Software Engineering

TU Delft, NL

- Designed research project proposals on LLM-driven software engineering
- Mentored 4 student teams through project implementation and academic paper writing
- Evaluated student papers and final presentations on technical implementation and research quality

Master Mentor

Sep. 2024 – Jan. 2025

MSc Computer Science

TU Delft, NL

- Guided 15 first-year students through their transition into the program during the Master Your Start event
- Held weekly meetings on specialization choice, study planning, research groups, and course selection
- Hosted social events to build community and connections among incoming students

Head Teaching Assistant

Feb. 2023 – Apr. 2023, Sep. 2024 – Nov. 2024

Software Architecture

TU Delft, NL

- Led the teaching assistant team for a 260-student course, coordinating student support
- Advised 10 student teams weekly on designing and documenting architectures for socially relevant systems
- Assessed essays, proof-of-concept implementations, and final project presentations
- Managed the development and communication infrastructure for all 65 project groups

SEMINARS & SUMMER SCHOOLS

- [Dagstuhl Seminar on Evaluation of AI Models in Software Engineering](#) '26, Schloss Dagstuhl, Germany
- [Machine Learning Summer School](#) '26, hosted by Columbia University and Bloomberg, New York, USA
- [Machine Learning Summer School](#) '26, hosted by Maincode, Melbourne, Australia
- [SEN Symposium](#) '25 and [AIWare BootCamp](#) '25, Amsterdam, The Netherlands

INDUSTRY EXPERIENCE

Full-Stack Developer	Oct. 2021 – Jul. 2023
<i>Lunar Zebro</i>	<i>Delft, NL</i>
• Designed a ground segment web application for the MINAR IX mission with the University of Edinburgh • Developed a rover swarm control application with real-time sensor analysis, mapping, and object detection • Improved rover control efficiency by 80% through real-time sensor data and video streaming • Configured an asynchronous message broker to coordinate four lunar rovers simultaneously • Represented TU Delft at the International Astronautical Congress '22, Paris, as the only student rover team	
Software Engineer	Apr. 2022 – Jul. 2022
<i>Feedback-Analytics</i>	<i>The Hague, NL</i>
• Developed a word processor with form logic support, reducing report creation time by 30% • Integrated the editor with the company's survey platform to generate customized feedback reports • Created an automated reporting pipeline that renders client reports as PDFs and dispatches them via email	

PROJECTS

CNN-Based URL Phishing Detection	Apr. 2024 - Jun. 2024
• Built an end-to-end release engineering pipeline taking a phishing detection model from training to production • Implemented data and model versioning, infrastructure provisioning, and containerized deployment • Set up automated testing and monitoring to track model quality and performance over time	
RAG for Complex QA	Jan. 2024 - Apr. 2024
• Developed a RAG pipeline using Contriever for retrieval and a quantized Llama 3 model for generation • Evaluated the pipeline on 2WikiMultiHopQA to study how context type affects multi-hop answer generation • Compared negative and random contexts against relevant retrieved passages, measuring their effect on answer quality	
Conversational Movie Agent	Nov. 2023 - Jan. 2024
• Built a movie recommendation agent with conversational memory on the Furhat social robot platform • Designed the agent's long- and short-term memory and integrated entity recognition and sentiment analysis • Ran a 50-participant user study showing that memory improved user satisfaction and recommendation accuracy	
Prompt Augmentation for Code Generation	Sep. 2023 - Nov. 2023
• Investigated whether partial software artifacts from the development context can steer LLMs toward correct code • Benchmarked WizardCoder and CodeGPT on MBPP, comparing augmentation strategies across model sizes • Raised pass@1 by 43% through test case augmentation, with the largest gains observed in bigger models	

TECHNICAL SKILLS & LANGUAGES

Programming Languages: Python, Java, C++, JavaScript, Scala, SQL

Technologies & Frameworks: Git, Docker, AWS, Hugging Face, PyTorch, Scikit-learn, Apache Spark, Apache Kafka

Languages Spoken: Romanian (Native), English (C2), German (A2)